

HEALTHCONNECT EXPERIENCE LAB

CROSS-TRACK INTEGRATION EVIDENCE

Data Analytics × Data Science | Week 6

Purpose — Document traceable evidence that cross-track collaboration produced analytical validation, challenge, refinement and a demonstrable change in the HealthConnect Week 6 decision-support output.

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Data Analytics Track | HealthConnect Pod 01

Cross-Track Collaborator

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Data Science Intern | HealthConnect Pod 01

1. Integration Objective & Collaboration Scope

The Week 6 cross-track objective was to compare validated Data Analytics evidence with Data Science modelling evidence, identify convergence and divergence, and determine which analytical conclusions should be confirmed, strengthened, nuanced, revised or rejected before translation into decision support.

Integration Requirement	Documented Response
Track involved	Data Science
Collaborator	AYDEN NGNINTEDEM DEMANOU — Data Science Intern, HealthConnect Pod 01
Dependency	Analytics required modelling evidence to challenge and refine the Week 6 evidence hierarchy.
Output provided by Analytics	Cross-Track Analytical Evidence Brief and KPI reference, including analytical population, priority signals, guardrails and modelling questions.
Output received from Data Science	Random Forest feature-relevance evidence, feature hierarchy and derived modelling features.
Integration activity	Cross-track review followed by an Analytics Integration Back assessment.
Demonstrable change	Booking Lead Time was strengthened; Previous No-Shows confirmed with guardrails; Distance and Age were nuanced; Reminder remained secondary/testable.

Integration Standard — Communication alone was not treated as integration. The collaboration was considered complete only after Data Science evidence changed or refined the Analytics interpretation.

2. Evidence A — Collaboration Initiated & Analytical Handoff


Ayden Demanou 8 h 29
 Hello Nancy


Ayden Demanou 8 h 29
 I'll be glad collaborating with you on this project. Let me know when we can have a meeting about it



 3 réponses Dernière réponse il y a 3 jours


Nancy Lee YIMBERE 14 h 46
 Hello Ayden,
 As discussed, I'm sharing the **Cross-Track Analytical Evidence Brief** from my Week 6 Data Analytics work on HealthConnect. It summarises the findings I have validated so far, particularly around **Booking Lead Time, Recorded Previous No-Shows, and Reminder Status**, as well as the combined-signal analysis. I have also included the specific areas where your **Data Science perspective and modelling evidence could challenge, confirm, nuance, or revise the analytical conclusions**.
 The aim is not simply to validate my findings, but to compare the Data Analytics evidence with your modelling results and identify what should ultimately be retained or refined for decision support.
 This should give us a clear basis for our discussion tomorrow at **3:00 PM WAT**.
 Thank you again for collaborating with me, Ayden. I really appreciate it.

Document Word


WK6_HealthConnect_Cross_Track_Analytical_Evi...
 Document Word

Evidence A1. Slack exchange documenting Ayden's agreement to collaborate and Nancy's structured pre-meeting analytical handoff.

The handoff explicitly positioned Data Science as a challenge and validation partner rather than as a confirmation step. The shared brief summarised the validated Analytics findings around Booking Lead Time, Recorded Previous No-Shows and Reminder Status and requested modelling evidence capable of confirming, challenging, nuancing or revising those conclusions.

modelling evidence could challenge, confirm, nuance or revise the analytical conclusions. The aim is not simply to validate my findings, but to cross-check evidence with your modelling results and identify what should ultimately be retained or refined for decision support. This should give us a clear basis for our discussion tomorrow at 3:00 PM WAT. Thank you again for collaborating with me, Ayden. I really appreciate it.

Document Word ▼



Jeudi 10 septembre ▼

Evidence A2. Traceable Word attachment: WK6_HealthConnect_Cross_Track_Analytical_Evidence_Brief, shared before the cross-track review.

3. Evidence B — Cross-Track Review Organised

HealthConnect Week 6 — Data Analytics × Data Science Cross-Track Review

Jeudi, 10 septembre · De 3:00 à 4:00pm

Fuseau horaire : Africa/Lagos

Informations de connexion Google Meet

Lien de l'appel vidéo : meet.google.com/vcp-xyzs-fgw

Meet

Real-time meetings by Google. Using your browser, share your video, desktop, and presentations with teammates and customers.

meet.google.com

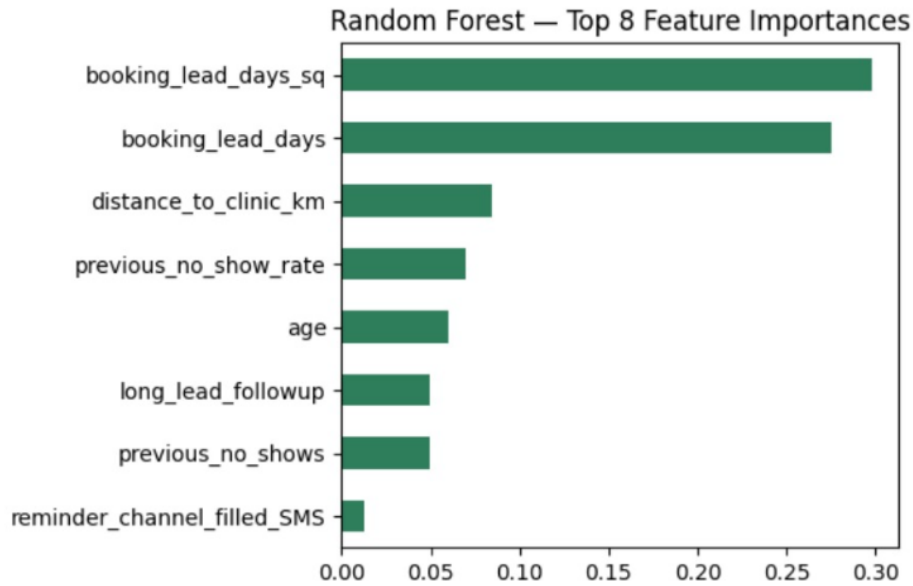


Evidence B1. Google Meet invitation for the HealthConnect Week 6 Data Analytics × Data Science Cross-Track Review, scheduled for 10 September, 3:00–4:00 PM WAT.

The review was organised around a defined analytical agenda: compare Analytics evidence with modelling evidence, examine feature relevance and limitations, and identify what should change in the final Week 6 decision-support interpretation.

Pre-Defined Review Questions — Does Booking Lead Time remain predictively useful in a broader feature set? Does Recorded Previous No-Shows add incremental information? Do modelling results support a complex interaction? How should Reminder Status be interpreted? Are Analytics underweighting any additional features?

4. Evidence C — Data Science Evidence Received



Evidence C1. Random Forest Top 8 Feature Importances returned from the Data Science work.

The later Random Forest evidence included both raw and engineered features. Booking Lead Time appeared through multiple representations (`booking\_lead\_days\_sq`, `booking\_lead\_days`, `long\_lead\_followup`), while recorded-history information appeared through `previous\_no\_show\_rate` and `previous\_no\_shows`. Distance to Clinic and continuous Age also appeared in the feature hierarchy.

Interpretation Guardrail — Feature importance is treated as predictive relevance, not causal importance. Importances of overlapping engineered features are not summed, and no causal mechanism is inferred from the Random Forest ranking.

5. Evidence D — Data Science Interpretation Received

High Importance Variables		
Variable Name	Type	Operational Reason for Inclusion & Model Relevance
1. <u>booking_lead_days</u>		
a.	Type: Numerical	
b.	Model Importance Score (RF): 0.346 (34.6%)	
c.	Most critical driver of non-attendance. Long gaps between scheduling and appointment dates severely erode patient intent and memory, exponentially increasing no-show risk.	
2. <u>distance_to_clinic_km</u>		
a.	Type: Numerical	
b.	Model Importance Score (RF): 0.138 (13.8%)	
c.	Strongest logistical friction factor. Longer travel distances create transportation barriers, increasing the likelihood of unannounced dropouts.	
3. <u>Age</u>		
a.	Type: Numerical	
b.	Model Importance Score (RF): 0.109 (10.9%)	
c.	Key demographic indicator. Younger adult patient cohorts show higher <u>volatility in attendance</u> <u>compared to older demographics</u> <u>who maintain rigid care routines</u> .	
4. <u>previous_no_shows</u>		
a.	Type: Numerical	
b.	Model Importance Score (RF): 0.069 (6.9%)	
c.	Primary behavioral predictor. Past unreliability is the strongest predictor of future unreliability. Retained as a raw count after removing collinear rate features	

Evidence D1. Data Science summary of high-importance variables supplied during the collaboration.

This artefact provided additional context on how Data Science interpreted the modelling variables. For the final Analytics output, causal-sounding statements in the source material were not adopted as causal conclusions. Instead, the evidence was translated into a more defensible distinction between predictive usefulness, observational analytical differentiation and causal evidence.

6. Integration Back — What Changed in Data Analytics

Analytics Finding Before Review	Data Science Input	Final Integration Decision
Booking Lead Time: strongest observed differentiation	Multiple lead-time representations appear prominently in the Random Forest feature hierarchy.	STRENGTHENED — retained as the primary prioritisation signal. Derived-feature importances are not added together.
Recorded Previous No-Shows: complementary supporting signal	Raw previous_no_shows and derived previous_no_show_rate retained in modelling.	CONFIRMED / NUANCED — useful complementary predictive information, subject to the patient-history data-quality guardrail.
Distance to Clinic: weak standalone association	Distance appears with greater relative predictive relevance in the multivariable model.	NUANCED — retained for multivariable evaluation, but not promoted to a standalone targeting rule.
Age Group: no meaningful standalone Analytics prioritisation	Continuous age appears in the Data Science feature hierarchy.	NUANCED — not a contradiction; continuous multivariable predictive utility may exist despite weak grouped univariate differentiation. No age-based decision rule.
Reminder Status: secondary contextual/diagnostic signal	Reminder-related feature appears low in the later Random Forest hierarchy.	CONSISTENT / TESTABLE — retain as secondary and evaluate reminder or attendance-support interventions through testing rather than causal inference.

Material Analytical Change — The collaboration did not simply reproduce the Week 5/Week 6 Analytics conclusions. It strengthened one priority signal, confirmed another with limitations, and changed the treatment of Distance and Age from purely

standalone Analytics findings to variables requiring multivariable interpretation.

7. Evidence-to-Decision Traceability

Cross-Track Evidence	Analytics Decision-Support Impact	Where Reflected
Lead-time feature relevance	Primary signal retained and strengthened; targeted attendance-support testing prioritised around longer booking lead times.	Advanced Analytics Notebook; DIAGNOSE & PRIORITIZE dashboard pages; Project Summary
Previous no-show history relevance	Complementary refinement signal retained with explicit longitudinal-quality guardrail.	Notebook; PRIORITIZE matrix; Project Summary
Distance predictive relevance	Standalone targeting rejected; variable retained for multivariable evaluation and Week 7 validation.	Notebook limitations/integration sections; Project Summary
Age predictive relevance	No demographic standalone rule introduced; predictive relevance distinguished from grouped univariate decision value.	Notebook integration/limitations; Project Summary
Reminder evidence	Reminder effectiveness not inferred; intervention testing recommended.	MONITOR decision signal; recommendations; Week 7 testing requirements

8. Contribution to HealthConnect

The Data Analytics × Data Science integration improved the HealthConnect decision-support framework in three ways. First, it increased confidence that Booking Lead Time is not only a robust descriptive signal but also remains relevant within a broader predictive feature set. Second, it confirmed the value of recorded no-show history while preserving the data-quality limitation that prevents overstatement. Third, it challenged Analytics to retain Distance and Age for multivariable evaluation without converting model relevance into unsupported operational or causal rules.

Final Cross-Track Outcome — Validated Analytics Evidence → Data Science Challenge → Predictive Evidence → Integration Back → Refined Decision Support → Week 7 Testing

9. Evidence Register

ID	Artefact	What It Demonstrates
A1	Slack collaboration exchange	Collaboration initiated and accepted; analytical handoff communicated.
A2	Cross-Track Analytical Evidence Brief	Structured Analytics output provided before review, including questions and guardrails.
B1	Google Meet invitation	Formal cross-track review organised around Data Analytics × Data Science.
C1	Random Forest Top 8 Feature Importances	Traceable technical modelling evidence received from Data Science.
D1	High Importance Variables summary	Data Science interpretation and feature-level feedback received.
E1	Week 6 Analytics Integration Back	Documented changes incorporated into the Notebook, Dashboard and Project Summary.

10. Methodological Guardrails

- Observational association and predictive relevance are not interpreted as causal effects.
- Random Forest feature importance is model- and feature-space-dependent; obsolete fixed percentages are not frozen after feature engineering.
- Overlapping engineered features are not summed to create an artificial importance score.

- Recorded patient-history variables remain subject to known longitudinal consistency limitations.
- The combined Analytics signals support targeted intervention testing, not deterministic individual no-show classification.
- Week 7 must validate operational usefulness and measured intervention outcomes before wider deployment.